

RealEstate Hub

Real Estate Management & Booking Platform

Project Idea Research & Planning

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1. Project Description

RealEstate Hub is a full-stack web application designed to simplify the process of finding, listing, and managing real estate properties.

The platform allows customers to search for properties, view detailed property information, and book appointments to visit properties. Real estate agents can add and manage their properties, upload property images and documents, and manage customer booking requests. Administrators have full control over users, agents, properties, and bookings.

The system solves the problem of managing real estate properties and customer requests manually by providing a centralized online platform that makes the process faster, easier, and more organized.

Target Users

- Customers looking for properties to buy or rent.
- Real estate agents who manage and advertise properties.
- Administrators who manage the entire platform.

Main Purpose

The main purpose of RealEstate Hub is to provide a secure and user-friendly platform for managing real estate properties, connecting customers with agents, and organizing property viewing appointments.

2. User Roles and Permissions

Admin

Permissions:

- Manage all users.
- Manage real estate agents.
- Manage all properties.
- Manage all bookings.
- View system statistics.
- Approve or remove properties.
- Delete inappropriate content.

Available Actions:

- Create users and agents.
 - View users, agents, properties, and bookings.
 - Update user and property information.
 - Delete users, agents, and properties.
 - Manage all booking requests.
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Real Estate Agent

Permissions:

- Manage their own profile.
- Create and manage their own properties.
- Upload property images.
- Upload property documents.
- View customer booking requests.
- Update booking status.

Available Actions:

- Add a new property.
 - View property details.
 - Edit property information.
 - Delete their own properties.
 - Upload images and documents.
 - Accept or reject booking requests.
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Customer

Permissions:

- Create and manage their own account.
- Browse available properties.
- Search and filter properties.
- View property details.
- Book property viewing appointments.
- View their bookings.
- Cancel their own bookings.
- Upload a profile picture.

Available Actions:

- Register and login.
- Search for properties.
- View property details.
- Create a booking.
- View booking status.
- Cancel a booking.
- Update profile information.

3. Main Features

Authentication Features

- User Registration.
- User Login.
- User Logout.
- Password Hashing.
- Forgot Password.
- Reset Password.
- Profile Management.
- Profile Image Upload.

Authorization Features

- Role-Based Access Control (RBAC).
 - Protected Routes.
 - Admin Dashboard.
 - Agent Dashboard.
 - Customer Dashboard.
 - Different permissions for Admin, Agent, and Customer.
 - Users can only access pages and actions allowed for their assigned role.
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Property Features

- Browse available properties.
 - Search properties.
 - Filter properties by location, price, and property type.
 - View property details.
 - Add new properties.
 - Edit property information.
 - Delete properties.
 - Upload multiple property images.
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Booking Features

- Book property viewing appointments.
 - View booking details.
 - Reschedule bookings.
 - Cancel bookings.
 - Accept or reject booking requests.
 - Track booking status.
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User Features

- Manage user profiles.
- Upload profile pictures.
- Update personal information.
- Manage users through the Admin Dashboard.

4. CRUD Operations

CRUD stands for Create, Read, Update, and Delete.

4.1 Property Management

Create

- Agents can add new properties.
- Admin can add properties.

Read

- Customers can view all available properties.
- Customers can view property details.
- Agents can view their own properties.
- Admin can view all properties.

Update

- Agents can edit their own property information.
- Admin can edit any property.

Delete

- Agents can delete their own properties.
- Admin can delete any property.

4.2 User Management

Create

- Users can create an account through registration.
- Admin can create users or agents.

Read

- Admin can view all users.
- Users can view their own profiles.

Update

- Users can update their own profiles. •
- Admin can update user information.

Delete

- Admin can delete user accounts.
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4.3 Booking Management

Create

- Customers can create property viewing bookings.

Read

- Customers can view their own bookings.
- Agents can view bookings for their properties.
- Admin can view all bookings.

Update

- Agents can accept or reject booking requests.
- Customers can reschedule their bookings.

Delete

- Customers can cancel their bookings.
 - Admin can delete bookings.
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4.4 Agent Management

Create

- Admin can create agent accounts.

Read

- Admin can view all agents.
- Customers can view agent information.

Update

- Agents can update their own profiles.
- Admin can update agent information.

Delete

- Admin can delete agent accounts.

5. Image and File Upload Features

5.1 User Profile Image

Allowed File Types:

- JPG
- JPEG
- PNG

Maximum Size:

- 5 MB

Uploaded By:

- Admin
 - Real Estate Agent
 - Customer
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5.2 Property Images

Allowed File Types:

- JPG
- JPEG
- PNG
- WEBP

Maximum Size:

- 10 MB per image
- Uploaded By:**
- Real Estate Agent
 - Admin

Purpose:

Agents can upload multiple images to display the property and help customers view its details before booking.

5.3 Property Documents

Allowed File Types:

- PDF
- JPG
- PNG

Maximum Size:

- 10 MB

Uploaded By:

- Real Estate Agent
- Admin

Purpose:

Agents can upload important property documents for administrative verification and management.

6. Database Management

The application will use a database to store and manage users, properties, agents, and bookings.

Users Table

- ID
 - Name
 - Email
 - Password
 - Role
 - Profile Image
 - Phone Number
 - Created At
- Roles:**
- Admin

- Agent
- Customer

Properties Table

- ID
- Title
- Description
- Price
- Location
- Property Type
- Purpose
- Bedrooms
- Bathrooms
- Area
- Images
- Documents
- Agent ID
- Created At

Bookings Table

- ID
- Property ID
- Customer ID
- Agent ID
- Date
- Time
- Status
- Created At

Booking Status:

- Pending
- Accepted
- Rejected
- Cancelled
- Completed

Database Relationships

- One Agent can manage many Properties.
- One Customer can create many Bookings.
- One Property can have many Bookings.
- Each Booking belongs to one Customer and one Property.
- <https://canva.link/9ilcf85s68sz0dq>